

Assignment - 4

①

Ambient Reflection	Diffuse Reflection	Specular Reflection
① Ambient light is result of indirect illumination ② $I_a = I_a k_a$ ③ overall background light.	① Result of irregular reflection. ② $I = I_p k_d \max\{\hat{L} \cdot \hat{N}, 0\}$ ③ Light is scattered in all directions from a rough surface.	① Result of regular reflection. ② $I = I_p k_s \max\{(RV)^n, 0\}$ ③ Shiny highlights where light is reflected in a preferred direction.

② $P = (6, 3, 4)$

$I_a = 4$

$k_a = 0.3$

$k_d = 0.6$

$k_s = 0.15$

$L = (1, 1, 25)$

$I_p = 12$

$V = (8, 8, 8)$

③ $n = 30$

$$d = \sqrt{(1-6)^2 + (1-3)^2 + (25-4)^2}$$

$$= \sqrt{470}$$

$$f_{att} = \max\left(1 - \left(\frac{d}{r}\right)^2, 0\right)$$

$$= 0.478$$

Ans:

$$|\vec{L}| = \sqrt{470}$$

(b)

$$\vec{L} = L - P$$

$$= (-5, -2, 21)$$

$$\hat{L} = \frac{-5}{\sqrt{470}}, \frac{-2}{\sqrt{470}}, \frac{21}{\sqrt{470}}$$

Since YZ plane, $\hat{n} = -\hat{i}$

$$\vec{R} = 2(\hat{L} \cdot \hat{n}) \hat{n} - \hat{L}$$

$$= 2 \times \left\{ \left(\frac{-5}{\sqrt{470}}, \frac{-2}{\sqrt{470}}, \frac{21}{\sqrt{470}} \right) \cdot \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \right\} \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} -5 \\ -2 \\ 21 \end{pmatrix}$$

$$= \left(\frac{15}{\sqrt{470}}, \frac{2}{\sqrt{470}}, -\frac{21}{\sqrt{470}} \right)$$

$$\vec{V} = (8, 8, 8) - (6, 3, 4)$$

$$= (2, 5, 4)$$

$$\hat{V} = \frac{2}{\sqrt{45}}, \frac{5}{\sqrt{45}}, \frac{4}{\sqrt{45}}$$

$$R.V = \frac{-41}{15\sqrt{94}}$$

$$n = 50$$

$$I_{total} = I_{aka} + I_{pfat} (k_d \max(\hat{L} \cdot \hat{n}, 0) + k_s \max(\hat{R} \cdot \hat{V}, 0)^n)$$

$$= 4 \times 0.3 + 12 \times 0.478 \times \left[0.6 \times \frac{5}{\sqrt{470}} \right] + 0.15 \times \left(\frac{-41}{15\sqrt{94}} \right)^{50}$$

$$= 1.99$$

Ans.

(b) - Gouraud Shading.

Explain interpolation concept, with the mathematical expressions.